

Nethmi Rajapaksha

📞 +94 77 601 2539 | ✉ nethmirajapaksha038@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio |
📍 Kelaniya, Sri Lanka

SUMMARY

Software Engineering undergraduate at the University of Kelaniya with hands-on experience building responsive web applications, full-stack systems, and AI-powered solutions using React.js, Next.js, Node.js, Express.js, and MongoDB. Experienced in developing REST APIs, authentication systems, responsive user interfaces, and AI integrations including RAG pipelines, vector search, Groq, and Google Gemini. Interested in Frontend, UI/UX, Full-Stack, and MERN development, with a focus on building practical, scalable, and user-friendly software solutions.

SKILLS

Programming Languages: JavaScript, TypeScript, Java, PHP

Frontend: React.js, Next.js, React Native, HTML5, CSS3, Tailwind CSS, Vite, React Router, Framer Motion

Backend: Node.js, Express.js, RESTful APIs, GitHub API (Octokit)

Databases: MongoDB, Mongoose, MySQL, PostgreSQL

AI / LLM Integration: RAG Pipelines, Embeddings, Vector Search, Groq API, Google Gemini API, Streaming Responses

Authentication & Security: JWT, bcrypt, Role-Based Access Control (RBAC)

Cloud & DevOps: AWS, Docker, GitHub Actions, CI/CD

Tools & Practices: Git, GitHub, Postman, Figma, VS Code, Agile Development, Responsive Design, Component-Based Architecture

PROJECTS

FurniHub — AI-Powered Furniture E-Commerce Platform (MERN) | [GitHub](#) Individual Project

- Developed a full-stack furniture e-commerce platform with product browsing, cart, wishlist, checkout, and order tracking.
- Implemented JWT authentication, bcrypt password hashing, and role-based access control for customers and administrators.
- Built an admin dashboard for product CRUD operations, order management, and sales statistics.
- Integrated Cloudinary for image management and Google Gemini for AI-powered room furniture recommendations.
- **Technologies:** React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT, Cloudinary, Google Gemini

ContentFlow AI — AI Content Generator SaaS | [GitHub](#) Individual Project

- Developed a full-stack AI content generation platform using Next.js, TypeScript, PostgreSQL, and Groq.
- Integrated Llama 3.3 70B Versatile through Groq to generate blog posts, articles, social media posts, emails, product descriptions, and captions.
- Implemented customizable content generation based on tone, length, keywords, and target audience.
- Built user authentication, content generation dashboard, searchable content history, account settings, and copy-to-clipboard functionality.
- Designed a responsive user interface with modern component-based architecture and dark mode.
- **Technologies:** Next.js, TypeScript, PostgreSQL, Groq, Llama 3.3, Clerk, Tailwind CSS

AI GitHub Codebase Assistant — RAG-Based Code Q&A Tool (Ongoing) | [GitHub](#) Individual Project

- Developing a web app where users paste a GitHub repo link and ask natural-language questions about its code.
- Implementing a RAG pipeline with the GitHub API, code chunking, embeddings, and vector search, using Groq to answer with file references.
- Building a Next.js chat interface with streaming responses and auto-generated documentation.
- **Technologies:** Next.js, TypeScript, Groq API, GitHub API (Octokit), Vector Search, Tailwind CSS

BookFair — Stall Reservation System | [GitHub](#)

Group Project | Jan 2026 – Apr 2026

- Developed and updated frontend pages for login, registration, home, stall mapping, vendor dashboard, and vendor profile using React and TypeScript.
- Implemented and maintained frontend types and stall-map utilities while integrating application screens with the project's API layer.
- Collaborated with the development team using GitHub and contributed to the frontend of a full-stack stall reservation platform.
- **Technologies:** React 18, TypeScript, Vite, Tailwind CSS, Bootstrap, TanStack Query, Axios, React Router, Spring Boot 3, Spring Security, JWT, Spring Data JPA/Hibernate, PostgreSQL

EDUCATION

University of Kelaniya, Sri Lanka

April 2024 – Present

BSc (Hons) in Software Engineering

Current GPA: 3.55 / 4.00

Relevant areas: Mobile Application Development, Net-Centric Web Applications, Health Informatics Applications

Bandaranayake Central College, Veyangoda, Sri Lanka

G.C.E. Advanced Level — Mathematics Stream

Combined Mathematics – A | Chemistry – B | Physics – B | Z-Score: 1.7594

LEADERSHIP & VOLUNTEERING

Head of Logistics IEEE Student Branch — University of Kelaniya

Member Software Engineering Students' Association (SESA) — University of Kelaniya

CERTIFICATIONS

- **CRUD Operations in MongoDB** — MongoDB
- **Agent Mode: Mission Accepted** — Postman Academy
- **Introduction to GitHub** — Microsoft Learn

LANGUAGES

English Professional Proficiency

Sinhala Native

REFERENCES

Prof. (Mrs) I.U. Hewapathirana

Professor, Software Engineering Teaching Unit
University of Kelaniya
ihewapathirana@kln.ac.lk
+94 77 732 9626

Dr. Nalin Warnajith

Senior Lecturer, Software Engineering Teaching Unit
University of Kelaniya
nwarnajith@gmail.com
+94 71 757 0140